



Jenkins

📌 Jenkins: The Ultimate Guide for DevOps Engineers 🚀

1 What is Jenkins?

Jenkins is an open-source automation server designed to streamline software development processes by implementing Continuous Integration (CI) and Continuous Deployment (CD). It automates repetitive tasks such as building, testing, and deploying applications, reducing manual effort and speeding up delivery cycles.

2 Why Jenkins in DevOps?

- ◆ Automates CI/CD pipelines – No manual intervention required.
- ◆ Enhances software quality – Detects bugs early, ensuring stability.
- ◆ Fast & Reliable Deployment – Automates rollbacks and deployments.
- ◆ Seamless integrations – Supports Git, Docker, Kubernetes, AWS, and more.
- ◆ Extensive plugin ecosystem – Over 1,800+ plugins for diverse automation tasks.
- ◆ Scalability – Runs distributed builds on multiple nodes for better performance.

3 Jenkins Architecture

Jenkins follows a master-agent architecture:

- 🖥️ Jenkins Master – Manages jobs, scheduling, and plugin configurations.
- 🖥️ Jenkins Agents (Slaves) – Execute build tasks, improving performance and distribution.

4 Key Features of Jenkins

- ✓ Pipeline as Code – Automate workflows using Jenkinsfile.
- ✓ Distributed Builds – Run jobs across multiple nodes.
- ✓ Declarative & Scripted Pipelines – Flexible pipeline configurations.
- ✓ Containerization Support – Works seamlessly with Docker & Kubernetes.
- ✓ Security & Access Control – Role-based access management.

5 Jenkins Pipeline: How it Works?

A Jenkins Pipeline is a script-based CI/CD workflow that defines the build and deployment process.

 Pipeline Stages:

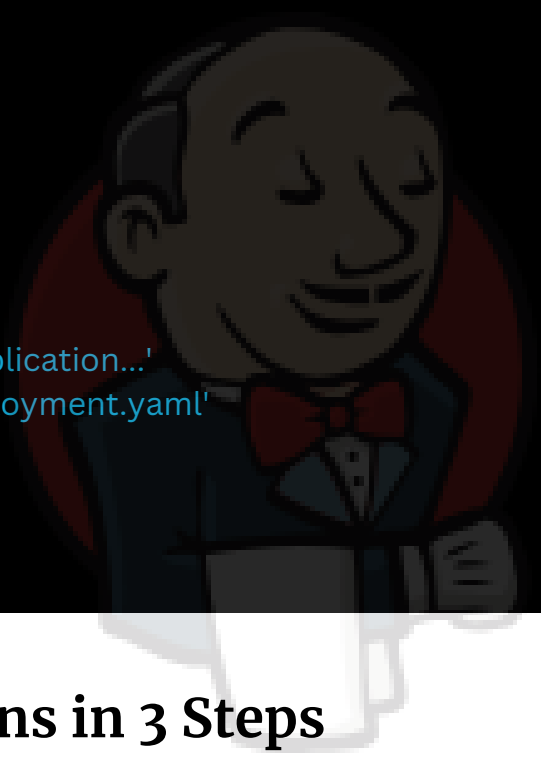
- ✓ Stage 1 – Code Commit: Developer pushes code to GitHub/GitLab.
- ✓ Stage 2 – Build: Jenkins fetches code and compiles the application.
- ✓ Stage 3 – Test: Runs unit & integration tests to catch bugs.
- ✓ Stage 4 – Deploy: Deploys the application to staging or production.

6 Common Jenkins Plugins

- 🔌 Git Plugin – Integrates with Git repositories.
- 🔌 Pipeline Plugin – Enables pipeline automation.
- 🔌 Docker Plugin – Supports containerized builds.
- 🔌 Kubernetes Plugin – Automates Kubernetes deployments.
- 🔌 SonarQube Plugin – Integrates code quality analysis.

7 Jenkinsfile Example (Declarative Pipeline)

```
pipeline {  
  agent any  
  stages {  
    stage('Build') {  
      steps {  
        echo 'Building the application...'  
        sh 'mvn clean install'  
      }  
    }  
    stage('Test') {  
      steps {  
        echo 'Running tests...'  
        sh 'mvn test'  
      }  
    }  
    stage('Deploy') {  
      steps {  
        echo 'Deploying the application...'  
        sh 'kubectl apply -f deployment.yaml'  
      }  
    }  
  }  
}
```



8 Setting Up Jenkins in 3 Steps

- ◆ Step 1: Install Jenkins (wget -q -O - <https://pkg.jenkins.io/debian/jenkins.io.key> | sudo apt-key add -)
- ◆ Step 2: Start Jenkins (sudo systemctl start jenkins)
- ◆ Step 3: Access Jenkins at <http://localhost:8080> and configure jobs.